

HITHA SHAMASUNDAR

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EDUCATION

Master of Science in Software Engineering

Arizona State University (GPA: 3.93/4.0)

May 2026

Tempe, AZ

Bachelor of Engineering in Computer Science and Engineering

NIE Institute of Technology - Visvesvaraya Technological University (CGPA: 8.18/10.0)

August 2023

Bengaluru, India

SKILLS

Programming Languages: Python, Java, JavaScript, C++

Methodology: Agile, SDLC, STLC, Waterfall

Web Technologies: HTML, CSS, TypeScript, Node JS, Next JS, React JS, JSON, Bootstrap, AJAX, Flask, Vue

API Development: REST API, GraphQL, Nest JS

Database: MySQL, PostgreSQL, NoSQL

DevOps: Docker, Git, Jenkins, CI/CD, GitHub, GitLab, Kubernetes

Testing tools & Automation frameworks: Postman, REST Assured, Selenium, Cucumber, Cypress, JUnit, TestNG, JMeter, Appium, Mocha & Chai

EXPERIENCE

Software Development Engineer in Test - Quality Assurance Engineer,

August 2023 - June 2024

Happiest Minds Technologies Limited

Bengaluru

- Developed **Cypress** automation test framework integrated with **Mocha and Chai** enabling asynchronous testing with custom hooks, assertions and test suites achieving a 35% reduction in flaky tests and improving maintainability.
- Leveraged automatic waits, time-travel debugging, and network stubbing capabilities to ensure robust UI test reliability and faster root-cause analysis for hAppl, a healthcare-focused conversational web chatbot.
- Performed extensive API automation testing using **REST Assured**, validating **JSON/XML payloads**, response codes, headers, and schema compliance improving API reliability and reducing manual effort by 50%.
- Engineered a custom browser automation framework using **Selenium WebDriver** integrated with **TestNG** to support test-driven development (TDD) for parameterized, data-driven, and modular testing workflows, improving regression coverage and accelerating test execution cycles by 30%.
- Implemented behavior-driven development (BDD) framework using **Cucumber** and **Gherkin** syntax, modularizing feature files and step definitions to align QA and development deliverables.
- Integrated **CI/CD** pipelines in **Jenkins** with automated test triggers via **Maven** and **Gradle**, while managing **Git** based version control, including branching, merging, and conflict resolution across multiple repositories.
- Collaborated in an **agile environment** to evaluate applications and systems for the Gen AI business unit using emerging technologies and AI-driven testing approaches & metrics, strengthening continuous quality validation.

Software Development Engineer Associate Intern

March 2023 - June 2023

HashedIn by Deloitte

Bengaluru

- Engineered a full-stack project management application using **Next.js** and **NestJS**, enabling users to create tasks, track issues, assign story points, and collaborate through role-based access, supporting 50+ concurrent project items across multiple user roles.
- Developed and maintained robust **RESTful API** endpoints in **NestJS** using a modular controller-service architecture with DTO-based validation, supporting CRUD operations for projects, tasks, and user management while improving API performance by 50%.
- Implemented a responsive frontend using **React Hooks** and **Next.js** routing, managing application state reducing client-side page reloads and improving navigation responsiveness by 35%.
- Designed relational data models using **Prisma ORM** with **PostgreSQL**, implementing schema migrations and optimized queries to support structured storage for projects, tasks, users, and story points across multiple relational tables.
- Implemented **JWT-based authentication** and role-based authorization, securing API endpoints and enforcing access controls for project collaboration features across 3+ user permission levels.

PROJECTS

DocBot: Document-Based Chatbot

- Designed and validated an end-to-end RAG pipeline, wrote test cases for text chunking accuracy, FAISS similarity scoring, and embedding consistency across varied document types and query patterns.
- Validated semantic retrieval quality against ground-truth queries, ensuring responses stayed grounded in source content and reducing hallucinations via controlled prompt augmentation. Tested REST API responses on a Flask + LangChain backend using assertions for schema compliance, response grounding, and failure handling across edge-case and out-of-scope inputs.

Adaptive Virtual Assistant

- Validated end-to-end behavior of an adaptive AI chatbot (Flask + Gemini API) by designing test scenarios for context retention, response relevance, and edge cases like empty inputs, session breaks, and ambiguous queries.
- Built and tested a semantic image retrieval pipeline over 100K+ Wikimedia images using OpenCLIP and ChromaDB — wrote integration tests to verify cosine similarity ranking and response relevance, improving contextual accuracy by 35%.
- Stress-tested a high-throughput async ingestion pipeline under concurrent load using aiohttp, validating elimination of API rate-limit failures and confirming a 50% gain in processing efficiency across multi-threaded workflows.